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Piezoelectric Based Class-E Resonant Inverter for Driving Surface Dielectric Barrier Discharge Plasma

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***Abstract*—The increasing development of distributed renewable energy enables the possibility of plasma-assisted local nitrogen fixation process in farms and individual households for fertilizer production. But traditional plasma generation requires large equipment and high voltages. Variable plasma impedance also makes it difficult to drive. This paper discusses the method and advantages to make plasma generation system smaller by integrating piezoelectric resonator into the driving converter and using it to drive surface dielectric barrier discharge reactors. We investigate performance of different plasma reactor designs, and characterize plasma impedance variance. We also build a very thin piezoelectric-based push-pull class E resonant converter with wide-range load driving ability, ZVS soft switching, and high voltage gain. By adopting the piezoelectric resonator, the inverter exhibits higher efficiency. The reactors can be integrated into PCBs with the converter, which can drive down cost. In general, the presented prototype plasma generation system is compact, simple-to-operate, and energy-efficient.**

***Index Terms*—dielectric barrier discharge plasma, piezoelectric resonator, resonant inverter**

I. Introduction

The nitrogen fixation for fertilizer production is an integral component of agricultural production [1]. Currently the primary method of fertilizer production (Haber Bosch process) is a large greenhouse gas contributor and cannot be done locally [2], [3]. The transport of fertilizer to farms also leads to extra CO_2 and pollutant emission, higher costs, as well as explosion and toxicity danger [4]. An environmental-friendly, locally viable fertilizer production process is then needed. In response, Non-Thermal Plasma-assisted Nitrogen Fixation (NTPNF) helps the agricultural sector to meet these requirements. As the deployment of distributed renewable energy sources accelerates, particularly the proliferation of distributed photovoltaic systems, the plasma generation can be implemented in each agricultural establishment and individual household. This means local green fertilizer production is possible, as shown in Fig. 1. The main challenge in this process is the bulky size of the converter (inverter) and the plasma generator (reactor). Typically, plasma generation involves huge vacuum chambers and super-high voltage [5], [6], which usually means large transformers. These shortcomings impede further wide spread of the distributed nitrogen fixation stated above.

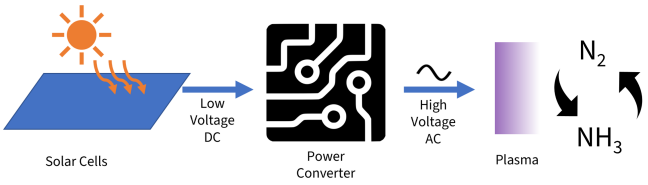


Fig. 1. Distributed plasma-assisted nitrogen fixation process in individual farms.

In this paper, we explore the possibility of miniaturizing the power converter and plasma reactor used in Fig. 1. To achieve this, we use Surface Dielectric Barrier Discharge (SDBD) to generate the plasma, and piezoelectric resonators to replace inductors in the power converter, both of which greatly compress the volumes of corresponding components. We demonstrate this benefit by building an ultra-thin plasma system, where a piezoelectric-based radio frequency (RF) class E resonant inverter successfully drives SDBD reactors to generate plasma.

II. Analysis of SDBD Plasma Reactors

In our system, the thin SDBD reactor functions as the load of the converter. Also, it is the interface between electric side and plasma side. Hence, appropriate reactor designs and their characterization are of great significance.

A. SDBD plasma reactor design

Surface Dielectric Barrier Discharge reactor's structure is shown in Fig. 2. When sufficiently high AC voltage is applied to both electrodes, the ambient air breakdown and discharge forms between the exposed electrode and the dielectric board surface. The parallel plate-like structure suggests that the reactor mainly behaves as a capacitor. The considerations for designing the reactors should include: 1) low dielectric loss (ESR); 2) low breakdown voltage (the minimum applied AC voltage for plasma to strike); 3) appropriate capacitance for converter driving.

Most SDBD reactors are operated at 10 kHz - 100 kHz, and their breakdown voltage can be as high as 6-10 kV [7]. Higher frequency reduces breakdown voltage, and brings higher discharge rate [8], so we set the working frequency of

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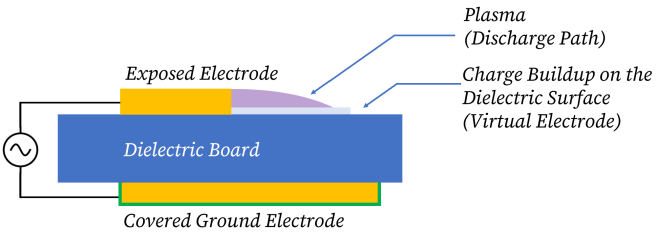


Fig. 2. Cross-section of an SDBD reactor.

our SDBD reactors to MHz. We then explore the dielectric material, thickness, and electrode geometry of the reactors.

We designed two series of reactors: 1) high frequency dielectric (Rogers/Teflon) with thickness of 510 μm/760 μm; 2) FR4 dielectric with thickness of 69 μm. The electrode geometry designs are shown in Fig. 3, where each cell is an individual reactor. Apart from dielectric material and thickness, these reactors differs in many other aspects, such as the exposed electrodes' geometry (shape: comb, needle, fan, tuning fork, grid, etc.; size), ground electrode's geometry (size; relative position to the exposed electrode), and treatment of sharp corners (rounded or not). All reactors are PCB-based, so their production is automated by PCB manufacturer and their quality is well controlled.

In high frequency scenarios, the Rogers/Teflon reactors have lower loss (higher quality factors) due to their lower dissipation factors (Df) than the FR4 reactors. However, the Rogers/Teflon boards' minimum thickness (limited by PCB manufacturing capabilities) hinders lowering of their breakdown voltage. On the other hand, FR4 dielectric can be made quite thin by designing a 4-layer board and adjusting the prepreg thickness between the 1st and the 2nd layer. For clearer comparison, Rogers reactor #1-16 and FR4 reactor #2-2, who share the same electrode geometry design, are compared in Table I. It explicitly shows the trade-off between efficiency (dielectric loss) and ease of generating plasma (breakdown voltage).

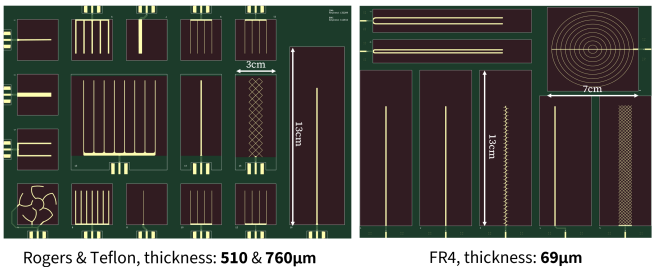


Fig. 3. Reactors designs based on different materials.

Electrode geometry design is also investigated via electromagnetic finite element analysis (FEA) simulation. We model the reactors' 3-D structure and simulate the electric field (E-field) in the vicinity of the exposed electrode, the region of interest (ROI) of air breakdown. When voltage of fixed amplitude is applied, higher E-field in the ROI indicates lower

TABLE I
COMPARISON BETWEEN ROGERS/TEFLON REACTOR #1-16 AND FR4 REACTOR #2-2 (SAME ELECTRODE GEOMETRY DESIGN).

Reactor design	#1-16	#2-2
Material	Rogers	FR4
Dielectric thickness	510 μm	69 μm
Quality factor@6.8 MHz	416	31.5
$V_{breakdown}$	2.2 kV	1 kV
Capacitance	12.1 pF	73.1 pF

breakdown voltage. For example, Fig. 4 shows the impact on E-field distribution of whether to offset the electrodes (reactor #2-2 and #2-4). Simulation result illustrates that the design with offset electrodes in Fig. 4(b) has higher E-field in the air near the exposed electrodes than that in Fig. 4(a). The comparison suggests that reactor #2-4 should have lower breakdown voltage, which is verified by experiments (1 kV vs. 860 V). The capacitance-conductance (C-G) FEA simulation also suggests that reactor #2-4 has lower dielectric loss due to the lower capacitance. Therefore, FEA simulation is proved to be helpful in optimizing electrode design of reactors.

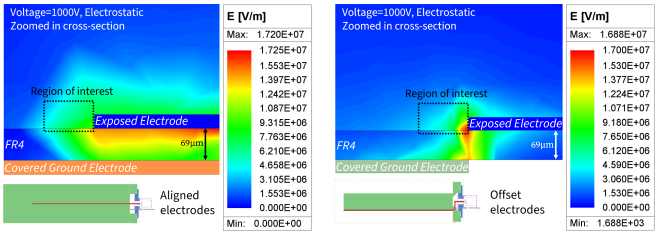


Fig. 4. E-field simulation of (a) reactor #2-2 (b) reactor #2-4.

B. Reactor driving and plasma impedance characterization

To drive these highly capacitive reactors, we initially use an RF power amplifier (PA), whose output impedance is 50 Ω. This necessitates the use of an impedance matching network. The experimental setup is shown in Fig. 5.

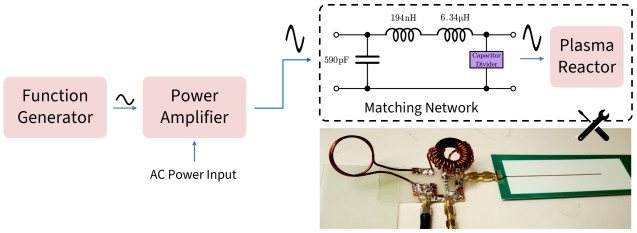


Fig. 5. Experimental setup of driving SDBD reactors by PA.

The matching network is designed to match the reactor #1-16 to 50 Ω at 13 MHz. For other reactors with different impedance, we can tune the frequency to ensure the PA's power delivery ability. Some photos of reactors generating plasma are captured in Fig. 6. Among these photos, discharge paths (streamer discharge) can be clearly observed. This denotes one key difference between kHz and MHz SDBD plasma [8]. Besides, different electrode's geometry induces different

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plasma patterns, and can fit in different reaction chamber shape. As the plasma keeps inducing heat and active ions, all reactors are operated in burst mode (300-1000 consecutive pulses every 10 ms) to avoid premature failure of the thin dielectric. The voltage waveform across the reactor during one burst (containing 600 pulses) is shown in Fig. 7. The voltage ramps up at the beginning of the burst until the voltage reaches the maximum and air breakdown takes place. Then, the changed load impedance of the plasma pulls down the voltage, and the reactor enters steady state.

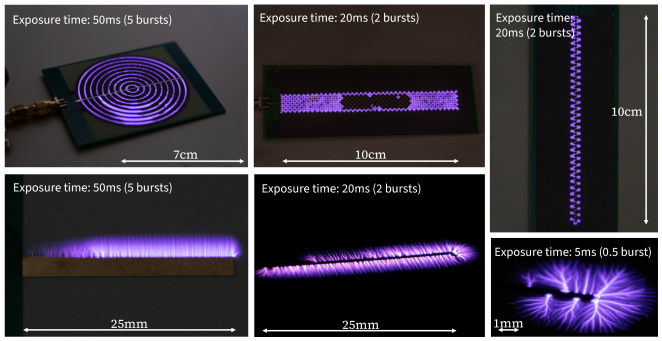


Fig. 6. Selected photographs of working reactors striking plasma.

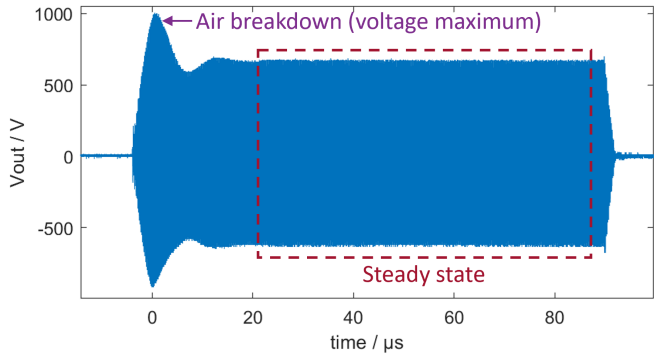


Fig. 7. Envelope of the voltage waveform across the reactor during one burst.

For more precise driving, it is necessary to characterize the impedance of the load, i.e., the SDBD plasma. An impedance model can be derived from SDBD reactor's structure [10], [11], as shown in Fig. 8. The variable components in the model stem from the "virtual electrode" formed by the deposited charge on the dielectric surface. The width of the virtual electrode varies as discharge develops. Accordingly, the total impedance of the plasma reactor fluctuates significantly.

By adding a current clamp probe and carefully calibrating the current measurement, we obtain the impedance variance with instantaneous power and phase, as in Fig. 9. The results are obtained from reactor #2-6, whose exposed electrode is in tuning fork shape, and the spacing between two prongs is 1.5 mm. Fig. 9(a) shows that as the instantaneous dissipated power increases, the equivalent serial C_p goes up while the equivalent serial R_p falls. This is due to a widened charge building up area (the virtual electrode) and increased discharge paths. In Fig. 9(b), both the voltage and current waveforms deviate from typical sinusoidal curves. Such divergence is caused by impedance variation within one period. Besides, during the positive half cycle, a voltage drop can be clearly observed, which is accounted for by the emerging discharge paths and, therefore, plummeted impedance. Such a highly variable impedance imposes the need for a wide-range driving converter, detailed in the next section.

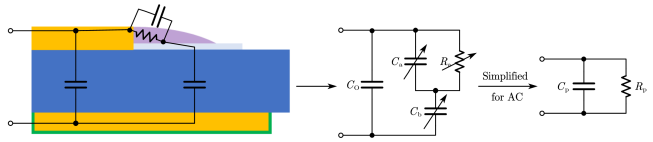


Fig. 8. SDBD plasma's impedance model.

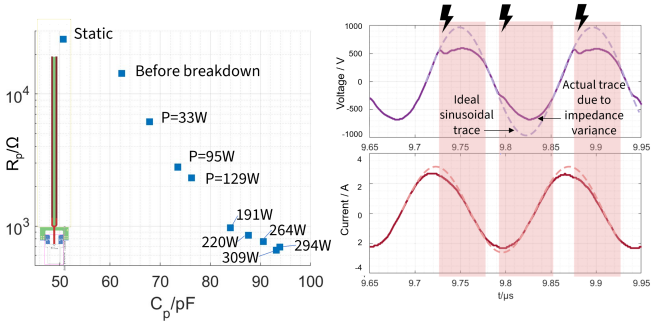


Fig. 9. Reactor #2-6's impedance variance with (a) instantaneous power (b) phase.

III. Design of Wide-Range Driving Converter

The core of the proposed plasma generation system is the power converter. It takes in low DC voltage, and drives the reactor with high AC voltage output. For the plasma generation application discussed in this paper, the driving converter should meet the following requirements: 1) high voltage gain (so that output voltage exceeds the breakdown voltage); 2) capability for driving a wide range of load impedance; 3) miniaturized size; 4) high efficiency.

To achieve this, we employ a two-step approach. First, we build a regular resonant inverter. Then, we replace the discrete inductors in the inverter with thin piezoelectric resonators to further reduce the converter's volume.

A. Converter design and experimental demonstration

The converter switching frequency is set to MHz. Higher switching frequency not only makes plasma generation easier as previously mentioned, but also shrinks the size of passive components. Nevertheless, the switching loss also increases. Therefore, we use class E resonant inverter, whose switching loss is minimal due to its soft switching nature, as the driving converter. Compared with the class D power amplifier, class E can also achieve high output voltage gain. The switches are implemented as SiC MOSFETs for their fast-switching

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ability and wide-bandgap benefits. Meanwhile, the resonant converter is quite sensitive to load variance to achieve zero voltage switching (ZVS) and zero voltage derivative (dv/dt) [14]. Generally, there are two approaches to deal with the load sensitivity: 1) small input inductance; 2) load variance compression. Classical class E converter includes a choke inductor with very high inductance, typically ranging from 10 μH to 100 μH and functioning as a constant DC current source. In contrast, the class E for variable load operation has a small inductor, usually in tens to hundreds of nH, as a resonant component and approximately an AC voltage source [12], [13]. However, the reduced input inductance lead to unacceptable high circulating and ripple current if outputting high voltage. The second approach is to build a resistance compression network (RCN) to compress the load variance [14]. But it mainly works for resistive variance, while our reactors show both resistive and capacitive change. Besides, the output voltage is divided by the RCN.

To compress the variance of the load impedance, we combine the two aforementioned ideas. We set the inductance of DC feed inductor to a relatively small value (4.7 μH, even smaller than resonant inductance in the output network), and add a shunt capacitor in the output end. Higher shunt capacitance achieves better load variance compression. However, this brings another two trade-offs: high capacitance paralleled at the output end 1) causes the switch drain nodes to see more capacitive load, which means more difficulty to achieve ZVS; 2) leads to higher circulating current that contributes to loss; 3) decrease total output impedance, thus lowering the output voltage gain. Therefore, we adopt push-pull topology to acquire the desired voltage gain and fine tune resonant tank parameters. The designed and built converter is shown in Fig. 10, whose component values are listed in Table II. The two SiC MOSFETs are switched in a complimentary manner, so that the push-pull topology can output nearly twice the voltage as a single class E converter does. In the built converter, the gate driver is implemented as two conventional gate driving ICs. However, at these high frequencies, SiC MOSFETs have high gating loss bacause of greater gate capacitance and higher turning-on V_{GS}. Therefore, a resonant gate driver can be potentially utilized to decrease loss further [15].

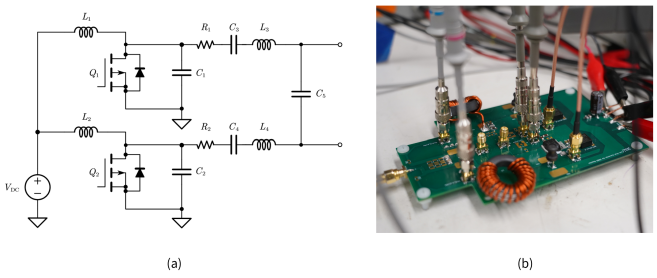


Fig. 10. (a) Designed push-pull class E resonant inverter; (b) built converter under test.

When driving different reactors with a wide range of load impedance, the built converter can both achieve ZVS and high voltage gain, as shown in Fig. 11. Here, both reactor #2-4 and #2-2 have a needle shape 100 mm $\times$ 1 mm exposed electrode (see the reactor structures in Fig. 4). However, #2-4's ground electrode are horizontally offset while #2-2's ground electrode are aligned. Thus reactor #2-2 exhibits much higher capacitance than #2-4. Despite the large difference in impedance between these two reactors (loads), the reactor can achieve high efficiency (by ZVS) and output a voltage higher than $V_{breakdown}$ for both of them.

TABLE II
KEY COMPONENT VALUES USED IN THE CONVERTER

Parameter	Value
Q_1, Q_2	C3M0120100J
Gate Driver IC	IXDN614YI
V_{DC}	0-200 V DC
L_1, L_2	4.7 μH
C_1, C_2	150 pF (C_{oss} included)
R_1, R_2	4 Ω (from ESR of parts)
C_3, C_4	315 pF
L_3, L_4	6.1 μH
C_5	18 pF
Switching frequency	6.8 MHz

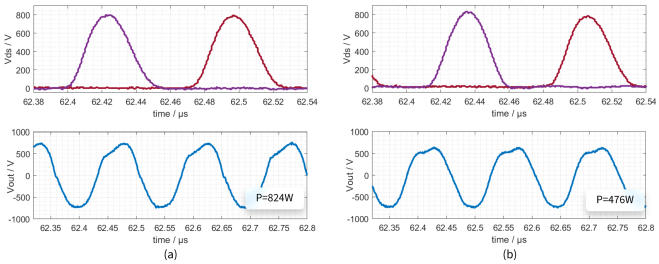


Fig. 11. SiC MOSFET drain-source ZVS and output voltage when driving (a) 29.3 pF reactor #2-4, V_{DC}=191 V; (b) 73.1 pF reactor #2-2, V_{DC}=175 V.

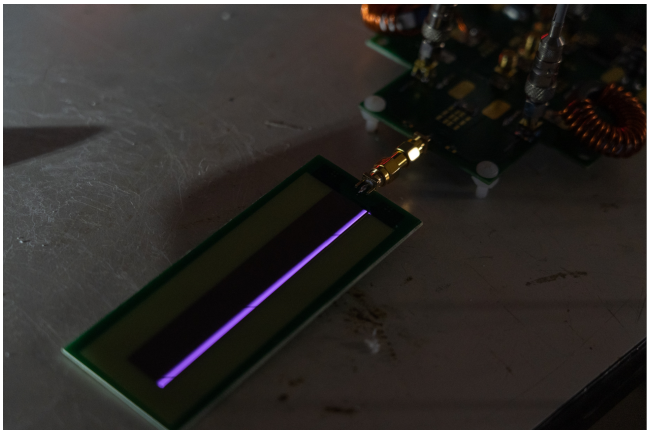


Fig. 12. Built converter driving reactor #2-4 generating plasma.

B. Adaptation into a piezoelectric-based converter

As shown above, there are two hand-wound inductors installed on the converter board. At high frequencies, it is

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difficult to optimize both the volume and the loss of an inductor simultaneously [16], [17]. The 68 ferrite material (NiZn) we used in the aforementioned converter achieves relatively good balance between the two trade-offs: smaller size compared with air-core inductors and lower loss at high frequency compared with other ferrite materials. Even so, these two inductors are still the largest components in the converter. The thickness of each inductor is about 13 mm, and with the leads (copper wires) included, the thickness exceeds 25 mm. The diameter is about 40 mm. In practical applications, if multiple converters are combined into a module, these inductors will occupy a significant amount of space, limiting the further miniaturization of the system. The volume of the fringing magnetic fields of these inductors is even larger. In addition, their ferrite cores lead to unsatisfying loss. At working frequency, these two inductors' ESR is measured to be 4 Ω, dissipating much energy.

To address this, we utilize Lithium Niobate ($LiNbO_3$) piezoelectric resonators as the resonant tank, replacing discrete inductors (L_3 and L_4) and capacitors (C_3 and C_4). Piezoelectric resonators efficiently absorb and release energy through the couple of mechanical vibration and electrical resonance. Therefore, piezoelectric resonators can function as an energy storing component, as well as a resonant element with high quality factor. Usually in a form of transparent thin planar component, and with a thickness of few hundred microns, piezoelectric resonators also possess huge advantage in volume reduction. These benefits make it very promising to substitute discrete inductors with piezoelectric resonators [18].

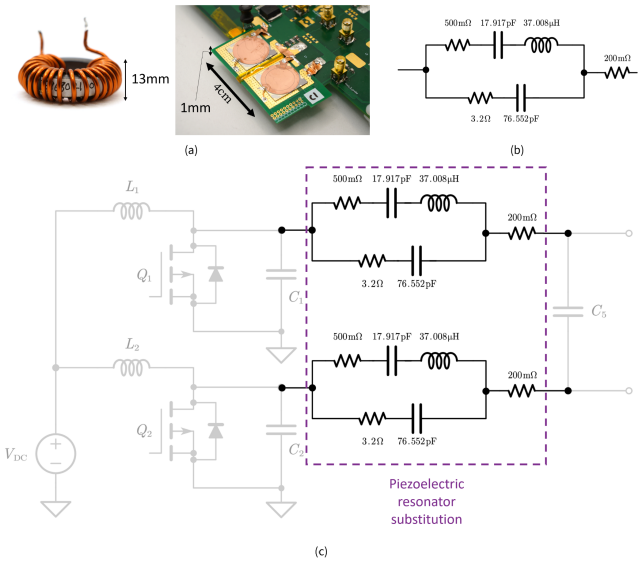


Fig. 13. (a) Size comparison of the hand-wound inductor and the piezoelectric resonators; (b) the BVD model of the piezoelectric resonator; (c) the circuit schematic with piezoelectric resonator adaptation.

In order to promote the voltage handling ability, we put two piezoelectric resonators in series to form one equivalent resonator. Previous work details the design and fabrication of the tested resonators [19], [20]. The installed piezoelectric equivalent Butterworth-Van Dyke (BVD) circuit model is shown in Fig. 13(b), where the equivalent component values are calculated based on the impedance acquired by a VNA analyzer. The RLC branch equivalently functions as a resonant tank with very high Q (quality factor). On the other hand, the capacitor branch, which is introduced by the capacitive coupling of the two electrodes of the piezoelectric resonator, allows for the passage of all high-frequency waves. This additional branch hence differentiate piezoelectric-based class E converters from traditional ones in terms of circuit behavior [21]. Therefore, we increase the output capacitance C_p from 18 pF to 68 pF to enhance the ability of filtering out unwanted high-frequency harmonics. Besides, the switching frequency is changed to 6.41 MHz to better fit these piezoelectric resonators. The adapted circuit schematic is presented in Fig. 13(c).

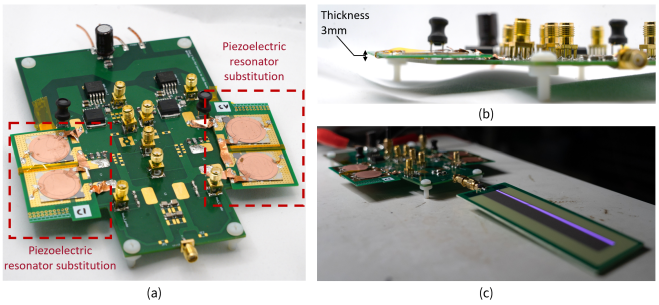


Fig. 14. (a) The built piezoelectric-based push-pull class E converter @ 6.41MHz; (b) side view of the piezoelectric resonator; (c) the working converter driving the reactor #2-4.

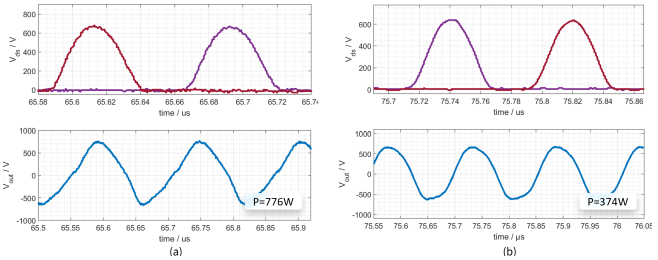


Fig. 15. V_{ds} and V_{out} waveform when driving (a) reactor #2-4, V_{DC}=180V; (b) reactor #2-2, V_{DC}=150V.

The converted piezoelectric-based push-pull class E resonant inverter is shown in Fig. 14. This adaptation greatly increases the converter's power density, reducing the thickness of converter's main part to 3 mm from over 13 mm. Also, the piezoelectric resonator offers higher efficiency due to its far higher quality factor ($Q_{piezoelectric}$=2130 vs. $Q_{hand\text{-}wound\ inductor}$=61.42). The converter can still achieve ZVS when driving different reactors, as shown in Fig. 15. The output voltage and voltage gain are presented in Fig. 16. The red and the purple traces represent voltage maximum during the whole burst and the amplification during the steady state (see Fig. 7), respectively. For the reactor #2-4, the converter remains a nearly constant voltage gain when input V_{DC} <30 V,

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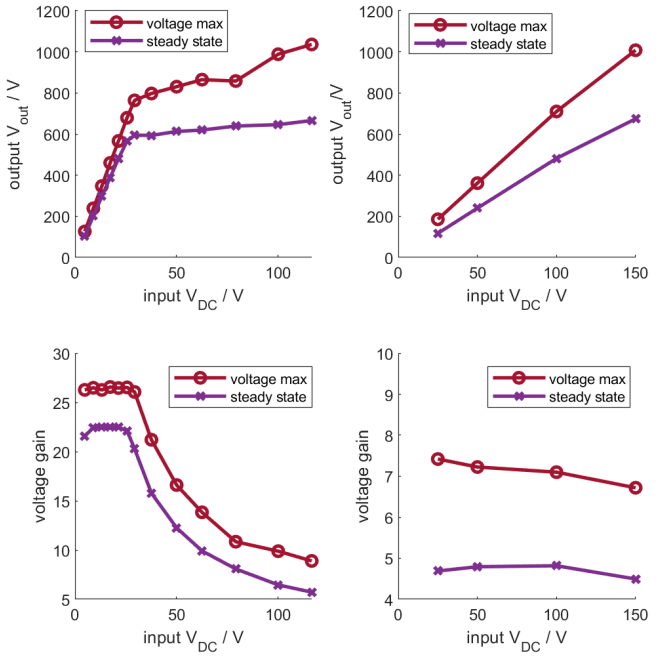


Fig. 16. The piezoelectric-based class E converter's output voltage maximum (red) and amplification during steady state (purple) by input V_{DC} for (a) reactor #2-4 (b) reactor #2-2.

i.e. when the there is no plasma. This characteristic holds until V_{out} exceeds the breakdown voltage. When V_{DC} >30 V ,i.e. when the plasma appears, there is just slight elevation in V_{out} as input V_{DC} increases. At this stage, the boosted input V_{DC} results in the increment of delivered power of the converter. Despite the sharp decrease in the reactor's impedance, leading to a subsequent reduction in the voltage gain of the converter after the plasma strikes, the converter achieves a maximum voltage gain greater than 5. For the reactor #2-2 as the load, because its capacitance is higher, the voltage gain becomes lower. Higher V_{DC} is needed to strike the plasma, so the voltage gain keeps stable in a wider range of V_{DC}. The overall voltage gain for #2-2 is also close to 5. Results in Fig. 16 demonstrates the ability of maintaining a high voltage gain for a wide range of load.

IV. Conclusions

This paper proposes the concept of a compact plasma generation system, and implements the idea by building a piezoelectric-based push-pull class E resonant inverter to drive SDBD plasma reactors. The main part of the converter is only 3 mm thick, and can output high AC voltage (>1000 V) to generate plasma with low DC voltage input in atmospheric conditions. These features make it easier to use, and more accessible to individual farms and households. For different applications, reactors are interchangeable because of the converter's ability to drive a wide range of load. By achieving ZVS, as well as utilizing high Q piezoelectric resonator, the built piezoelectric-based class E converter can function with high efficiency.

References

[1] M. Anowarul Islam and Albert Tetteh Adjesiwor, "Nitrogen Fixation and Transfer in Agricultural Production Systems," *InTech eBooks*, Feb 2018.

[2] C. Smith, A. K. Hill, and L. Torrente-Murciano, "Current and future role of Haber–Bosch ammonia in a carbon-free energy landscape," *Energy & Environmental Science*, vol. 13, no. 2, pp. 331–344, Feb 2020.

[3] J. W. Erisman, "How ammonia feeds and pollutes the world," *Science*, vol. 374, no. 6568, pp. 685–686, Nov 2021.

[4] T. P. Hignett, "Transportation and Storage of Ammonia," *Fertilizer Manual*. Dordrecht: Springer Netherlands, pp. 73–82, 1985.

[5] A. Bogaerts and E. C. Neyts, "Plasma Technology: An Emerging Technology for Energy Storage," *ACS Energy Letters*, vol. 3, no. 4, pp. 1013–1027, Mar 2018.

[6] C. Yong-Nong and K. Chih-Ming, "Design of Plasma Generator Driven by High-frequency High-voltage Power Supply," *Journal of Applied Research and Technology*, vol. 11, no. 2, pp. 225–234, Apr 2013.

[7] J. Kriegseis, B. Möller, S. Grundmann, and C. Tropea, "Capacitance and power consumption quantification of dielectric barrier discharge (DBD) plasma actuators," *Journal of Electrostatics*, vol. 69, no. 4, pp. 302–312, Aug 2011.

[8] W. Jiang, H. Wu, Z. Wang, L. Yi, and Y. Zhang, "Gas breakdown in radio-frequency field within MHz range: a review of the state of the art," *Plasma Science & Technology*, vol. 24, no. 12, pp. 124018–124018, Dec 2022.

[9] J. P. Dedrick, "Radio-frequency surface discharges and their application to active flow control," Ph.D. dissertation, Australian National University, Canberra, ACT, Australia, 2013.

[10] N. Wilde, H. Xu, N. Gomez-Vega, and Steven, "A model of surface dielectric barrier discharge power," *Applied Physics Letters*, vol. 118, no. 15, pp. 154102–154102, Apr 2021.

[11] J. Xu, Kawin Surakitbovorn, B. Wang, M. A. Cappelli, and J. Rivas-Davila, "Frequency-Selective MHz Power Amplifier for Dielectric Barrier Discharge Plasma Generation," *IEEE open journal of power electronics*, vol. 3, pp. 846–855, Jan 2022.

[12] Łukasz Rosłaniec, A. S. Jurkov, A. A. Bastami and D. J. Perreault, "Design of Single-Switch Inverters for Variable Resistance/Load Modulation Operation," *IEEE Transactions on Power Electronics*, vol. 30, no. 6, pp. 3200–3214, Jun 2015.

[13] X. Huang, Z. Yu, Y. Dou, S. Lin, Z. Ouyang, and Michael A.E. Andersen, "Load-Independent Push–Pull Class E^2 Topology With Coupled Inductors for MHz-WPT Applications," *IEEE Transactions on Power Electronics*, vol. 37, no. 7, pp. 8726–8737, Jul 2022.

[14] Y. Han, O. Leitermann, D. A. Jackson, J. M. Rivas, and D. J. Perreault, "Resistance Compression Networks for Radio-Frequency Power Conversion," *IEEE Transactions on Power Electronics*, vol. 22, no. 1, pp. 41–53, Jan 2007.

[15] Z. Ye, Z. Tong, L. Gu, and J. Rivas-Davila, "A High Frequency Resonant Gate Driver for SiC MOSFETs," in *2021 IEEE 22nd Workshop on Control and Modelling of Power Electronics (COMPEL)*, Nov 2021.

[16] C. R. Sullivan, B. A. Reese, Aaron, and Phyo Aung Kyaw, "On size and magnetics: Why small efficient power inductors are rare," in *2016 International Symposium on 3D Power Electronics Integration and Manufacturing (3D-PEIM)*, Jun 2016.

[17] P. Papamanolis, T. Guillod, F. Krismer, and J. W. Kolar, "Minimum Loss Operation and Optimal Design of High-Frequency Inductors for Defined Core and Litz Wire," *IEEE Open Journal of Power Electronics*, vol. 1, pp. 469–487, 2020.

[18] E. A. Stolt, W. D. Braun, C. Y. Daniel, and J. M. Rivas-Davila, "Piezoelectric Resonator Second Harmonic Cancellation in Class Φ2 Inverters," in *2021 IEEE 22nd Workshop on Control and Modelling of Power Electronics (COMPEL)*, Nov 2021.

[19] K. Nguyen et al., "Spurious-Free Lithium Niobate Bulk Acoustic Resonator for Piezoelectric Power Conversion," in *2023 Joint Conference of the European Frequency and Time Forum and IEEE International Frequency Control Symposium (EFTF/IFCS)*, May 2023.

[20] E. Stolt, W. Braun, K. Nguyen, Vakhtang Chulukhadze, R. Lu, and J. Rivas-Davila, "A Spurious-Free Piezoelectric Resonator Based 3.2 kW DC-DC Converter for EV On-Board Chargers," *IEEE Transactions on Power Electronics*, vol. 39, no. 2, pp. 2478–2488, Feb 2024.

[21] C. Daniel, E. Stolt, and J. Rivas-Davila, "Class E Power Amplifier with Piezoelectric Resonator Output Branch," in *2021 IEEE 22nd Workshop on Control and Modelling of Power Electronics (COMPEL)*, Nov 2021.

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